



Rajshahi University of Engineering & Technology

Department of Mechatronics Engineering

MTE LAB REPORT – 04

Course No.: **MTE 3202**

Course Title: **Power Electronics and Drives Sessional**

Submitted By:

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Experiment No.: 04

Name of the Experiment:

Observation of a Single-Phase SCR with R Load.

Objective:

1. To analyze the operation of a single-phase SCR circuit with a resistive load by observing its voltage and current waveforms.
2. To understand the effect of varying the firing angle on the conduction period and power delivered to the load.
3. To investigate the performance parameters of the circuit.

Theory:

A controlled single-phase SCR (Silicon Controlled Rectifier) operates by using a gate triggering pulse to control the conduction period in an AC circuit. The phase-control thyristor remains in a non-conducting state until a gate pulse is applied, at which point it turns on and conducts current until the AC voltage crosses zero and the current falls below the holding level and is turned off due to natural or line commutation. By adjusting the firing angle (the delay in triggering within each AC cycle), the average power delivered to the load can be controlled. This principle is essential in phase-controlled rectifiers, where partial conduction of the AC waveform allows precise voltage and power regulation. Such control is widely used in applications like motor speed regulation, heater control, and dimming circuits [2].

Effect of Firing Angle (α):

- For $\alpha = 0^\circ$, the circuit behaves like a normal diode rectifier.
- For $\alpha = 90^\circ$ or more, the output voltage is significantly reduced, and for $\alpha \geq 180^\circ$, no conduction occurs.

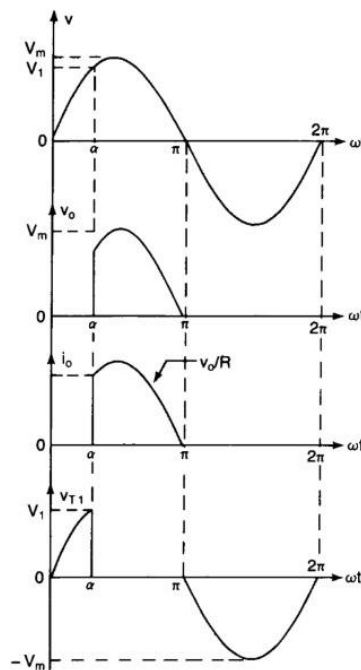


Figure 4.1: General Waveform of Single-Phase Semi-Converter SCR with R Load [1]

Required Apparatus:

1. Power Electronics Board (DE LORENZO)
2. Oscilloscope (Protek 5100)

Circuit Diagram:

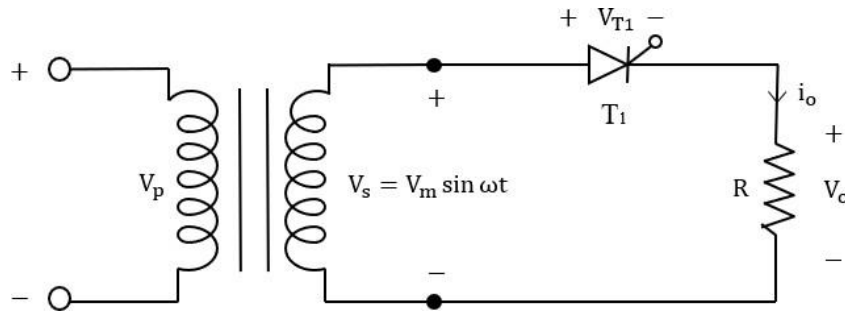


Figure 4.2: Circuit Diagram of a Single-Phase Semi-Converter SCR with Resistive Load

Procedure:

1. The circuit was assembled on the power electronics board according to the circuit diagram (Figure 4.2).
2. Next, the input and output parameters were measured and recorded in the data table using the oscilloscope.
3. Finally, the relevant waveforms were captured from the oscilloscope.

Data table and calculation:

SI No.	Case	Firing Angle, α	V_m (V)	V_{dc} (V)		V_{rms} (V)		Frequency (Hz)
				Theory	Measured	Theory	Measured	
01	R Load	45°	10.4	2.8	2.826	4.8	4.96	46.98
Performance Rectification, η				Form Factor, FF		Ripple Factor, RF		
32.46%				1.76		1.45		

Peak Input Voltage, $V_m = 10.4$ V

$$\begin{aligned}\text{Average Output Voltage, } V_{dc} &= \int_0^T \frac{1}{T} V_{out}(t) d\omega t \\ &= \frac{V_m}{2\pi} (1 + \cos \alpha)\end{aligned}$$

$$= \frac{10.4 (1 + \cos 45^\circ)}{2\pi}$$

$$= 2.8 \text{ V}$$

$$\text{RMS value of output voltage, } V_{\text{rms}} = \sqrt{\int_0^T \frac{1}{T} V_{\text{out}}^2(t) d\omega t}$$

$$= 4.8 \text{ V}$$

$$\text{FF} = \frac{V_{\text{rms}}}{V_{\text{dc}}}$$

$$= 1.76$$

$$\text{RF} = \sqrt{\text{FF}^2 - 1}$$

$$= 1.45$$

Performance Rectification, $\eta = 32.46\%$

Discussion:

In the experiment, the operation of a single-phase SCR with an R load was observed by analyzing the voltage and current waveforms. The SCR was triggered at different firing angles, and its conduction characteristics were examined. It was found that as the firing angle increased, the conduction period decreased, reducing the power delivered to the load. The waveforms confirmed the expected behavior, demonstrating phase control of the AC signal. Some minor deviations were observed due to circuit losses and triggering inaccuracies.

Conclusion:

The principle of controlled rectification using a single-phase SCR with an R load was successfully demonstrated. The relationship between the firing angle and power delivered to the load was verified through waveform analysis. It was concluded that increasing the firing angle reduces the conduction period, thereby lowering the output power. The experiment validated the theoretical concepts of SCR phase control. Overall, the results confirmed the effectiveness of SCRs in AC power regulation applications. This converter is not normally used in industrial applications because it has high ripple content and low ripple frequency.

References:

- [1] M. H. Rashid, *Power Electronics: Circuits, Devices, and Applications*, 4th ed., Pearson, 2013.
- [2] R. Boylestad and L. Nashelsky, *Electronic Devices and Circuit Theory*, 11th ed., Pearson, 2015